

Product Forms in Closed Order Independent Queues

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ABSTRACT

There is a large literature on product form queueing models, but it is overwhelmingly focused on open systems. However, much less is known about product forms in closed queues, and existing results typically apply only to restrictive special cases. In this paper, we introduce the first broad family of closed queues with stationary product form distributions, which we call *Front Index Order Independent* (FIOI) queues. Closed models are of interest both as models of genuinely closed systems and as closed counterparts of open systems. In the latter case they are important because the throughput of the closed system determines the stability threshold of the corresponding open model.

Our first contribution is to identify this general FIOI family and derive its associated product form, in which the stationary probability of a state factors into a simple product of probabilities associated with the jobs present in the system. The basic model assumes uniform completion probabilities among jobs in service, and we then extend the framework in four directions: allowing the class of the arriving job to depend on the departing job; allowing completion probabilities to depend on position in the queue; allowing countably infinite class spaces; and allowing more general closed systems in which a completion may trigger multiple arrivals and the number of jobs in the system need not remain constant across states.

We then formulate *Front Sum Order Independence* (FSOI), a closed model analogue of assumptions from the open order independent queueing literature, and derive a corresponding product form for the closed system paralleling the open FSOI result.

1 Introduction

In this paper, we introduce and study a general family of closed FCFS queues with product form stationary distributions, that is, systems with no exogenous arrivals in which each completion triggers a new arrival. Closed systems are important both in their own right and as closed counterparts of open models. A concrete example is a *constant work-in-process* (CONWIP) production system, in which a fixed number of jobs are maintained on the line at all times: when one job completes service and exits, a new job is immediately released into the system. This type of pull-based control is common in manufacturing. For instance, one may view a semiconductor fabrication facility producing many

different chip types as operating with a fixed number of lots in process, where the completion of one lot triggers the release of another [5].

Closed models are also important because the throughput of the closed system determines the stability threshold of the corresponding open system. Such a closed system is often called the saturated model. The connection between saturated and open systems is a folk theorem first formalized by Baccelli and Foss [1].

The key structural condition in this paper is *Front Index Order Independence* (FIOI). FIOI requires that the set of in-service jobs depend only on the *multiset* of classes at the front (earliest jobs in FCFS order) of the queue, and not on their internal order. It can also depend on the specific ordering of the jobs at the back of the queue.

One source of FIOI models are the Multiserver-job models: in these models, a single job may require multiple servers simultaneously for the duration of its service, a setting that has received increasing attention because modern compute clusters routinely run such jobs [3]. More generally, the same idea extends to multiresource-job systems, where feasibility is governed by multidimensional resource constraints. It also applies to other closed FCFS systems, including processor-sharing-type models in which service is shared among several front jobs, as well as other service rules. For example, one may consider a queue with red, blue, and green jobs, in which the server always serves the maximal front segment containing jobs of at most two different colors.

We begin by extending the single-duration saturated MSJ model studied in Grosof et al. [2], in which all jobs have identical exponential service durations. In that setting, the embedded Markov chain admits a simple product form stationary distribution. Our first contribution is to generalize this result to a basic closed queueing model satisfying FIOI and to show that it also admits a product form stationary distribution. In particular, the MSJ product form arises because the MSJ service rule satisfies FIOI, so the single-duration MSJ model is a special case of our general framework.

Our next contribution is to show that this structure persists for a broad family of closed models satisfying FIOI. Under FIOI, we then extend the framework in four directions: allowing the class of the arriving job to depend on the departing job; allowing completion probabilities to depend on position in the queue; allowing countably infinite class spaces; and allowing more general closed systems in which a completion may trigger multiple arrivals and the number of jobs in the system need not remain constant across states.

Our final contribution is a product form result more di-

rectly inspired by the open order independence (OI) literature; see [4]. We adapt the assumptions of Krzesinski [4] to the closed model setting and call the resulting condition *Front Sum Order Independence* (FSOI). FSOI requires that the cumulative completion rate of the first j positions depend only on the multiset of the first j classes. Under this condition, we obtain a more complicated product form similar to the open FSOI product form for our closed model.

The remainder of the paper is organized as follows. **Section 2** introduces the basic closed queueing model, recalls the single-duration saturated MSJ example and its product form result from [2], and then introduces FIOI and derives our order-independent product form. **Section 3** presents the FIOI framework and the four extensions listed above. **Section 4** develops the FSOI framework, and derives a product form.

2 Model

We begin with a general fixed-length closed FCFS queue. The system contains exactly k jobs at all times, ordered by FCFS. When a job completes service, it departs and a new job immediately arrives and joins the back of the queue. Each job has a class in some class set $\mathcal{C}$ and successive arrivals are i.i.d. with class probabilities $(p_c)_{c \in \mathcal{C}}$.

The state is an ordered vector $m = (m_1, \dots, m_k) \in \mathcal{C}^k$, where m_i denotes the class of the i th job in FCFS order. Let $s : \mathcal{C}^k \rightarrow \{1, \dots, k\}$ denote the number of in-service jobs in a state. In state $m = (m_1, \dots, m_k)$, the next completion occurs *uniformly* from among positions $\{1, \dots, s(m)\}$.

One may study either the continuous-time Markov chain or the embedded discrete-time Markov chain that updates once per completion. These two chains are closely related, and the stationary distribution of one determines that of the other. In this paper, we focus on the embedded DTMC.

We next give an example, due to Grosf et al. [2], which motivates the broader framework developed in the remainder of the paper.

2.1 Single-duration MSJ

Consider the saturated multiserver-job (MSJ) model in the single-duration setting studied by Grosf et al. [2]. There are k servers and exactly k jobs in the system. Successive arrivals are i.i.d., with class probabilities $(p_c)_{c \in \mathcal{C}}$. All jobs have i.i.d. service durations $\text{Exp}(\mu)$, independent of class.

Define $\sigma(m) := \max \left\{ r \in \{1, \dots, k\} : \sum_{j=1}^r m_j \leq k \right\}$. Then positions $\{1, \dots, \sigma(m)\}$ are in service, while positions $\{\sigma(m)+1, \dots, k\}$ wait in queue. Since all service rates are equal, the next completion occurs uniformly among the jobs currently in service, that is, among the first $\sigma(m)$ positions. Therefore, this model is a special case of the general fixed-length closed queue introduced above, with $s(m) = \sigma(m)$.

THEOREM 1 ([2]). *In the single-duration model, $\pi(m) = \prod_{j=1}^k p_{m_j}$, $m \in \mathcal{C}^k$, is the stationary distribution of the embedded DTMC.*

2.2 Finite-Class FIOI Model

We now extend **Section 2.1** to a broader family of closed models satisfying the following assumption,

ASSUMPTION 1 (FIOI). *For any $m, m' \in \mathcal{C}^k$, if the first $s(m)$ entries of m' are a permutation of the first $s(m)$ entries of m , and $m'_j = m_j, \forall j > s(m)$, then $s(m') = s(m)$.*

The assumption says that the value of $s(m)$ depends only on the multiset of classes occupying the first $s(m)$ positions of the queue, and not on their order, as well as specific sequence after $s(m)$. We call this *Front Index Order Independence* (FIOI). In **Theorem 1**, $s(m) = \sigma(m)$, so **Assumption 1** holds because the jobs in service are exactly the maximal front segment that fits within capacity, and this is unchanged by permuting those jobs. We next show that the product form in **Theorem 1** continues to hold under **Assumption 1**.

THEOREM 2. *Assume **Assumption 1**. Then $\pi(m) = \prod_{j=1}^k p_{m_j}$, $m \in \mathcal{C}^k$, is the stationary distribution of the embedded DTMC.*

Theorem 2 serves as the starting point for our later generalizations. In the next four subsections, we extend it in four directions.

3 FIOI Generalizations

In this section, we generalize the basic finite-class, fixed-length and i.i.d. arrival closed model introduced in **Section 2.2** under **Assumption 1**. We consider the four extensions outlined in the introduction.

3.1 Departure Dependent Arrival

We first generalize the i.i.d. arrival by allowing the class of the new job to depend on the class of the departing job, while assuming **Assumption 1**. Let $P = (P_{ab})_{a,b \in \mathcal{C}}$ be a dependency matrix where $P_{ab} \geq 0$ for all $a, b \in \mathcal{C}$ and $\sum_{b \in \mathcal{C}} P_{ab} = 1$ for all $a \in \mathcal{C}$. If the completed job has class a , then the arriving job has class b with probability P_{ab} . The product form continues to hold,

THEOREM 3. *Assume **Assumption 1**. Let $\nu = (\nu_a)_{a \in \mathcal{C}}$ be a stationary distribution of P . Then $\pi(m) = \prod_{j=1}^k \nu_{m_j}$, $m \in \mathcal{C}^k$, is a stationary distribution of the embedded DTMC.*

3.2 Position Dependent Departure

We next generalize the model from the beginning of **Section 2.2** by allowing an arbitrary state-dependent completion law. For each state $m \in \mathcal{C}^k$ and each position $j \in \{1, \dots, k\}$, let $\Delta_j(m) \in [0, 1]$ denote the probability that the job in position j is the next to complete.

The model of **Section 2.2** is the special case

$$\Delta_j(m) = \frac{1}{s(m)} \mathbf{1}\{j \leq s(m)\}.$$

ASSUMPTION 2 (FIOI- Δ). *For any $m, m' \in \mathcal{C}^k$, if the first $s(m)$ entries of m' are a permutation of the first $s(m)$ entries of m , and $m'_j = m_j, \forall j > s(m)$, then $\Delta_j(m') = \Delta_j(m)$ for all $j \leq s(m)$.*

We have the following theorem,

THEOREM 4. *Assume **Assumption 2**. Then $\pi(m) = \prod_{j=1}^k p_{m_j}$, $m \in \mathcal{C}^k$, is the stationary distribution of the embedded DTMC.*

3.3 Countably Infinite State Spaces

We next extend **Theorem 2** to the case where the class set $\mathcal{C}$ is countably infinite, while assuming **Assumption 1**. The state space remains $\mathcal{C}^k$, and arrivals are i.i.d. with class probabilities $(p_c)_{c \in \mathcal{C}}$.

A natural example is a MSJ model with fractional server demands in $[0, 1]$, where the class distribution is supported on a countable subset of $[0, 1]$. With modern supercomputers comprising many nodes, discrete server demands are naturally approximated by a nearly dense distribution on $[0, 1]$.

THEOREM 5. Assume [Assumption 1](#), and let $\mathcal{C}$ be countably infinite. Then $\pi(m) = \prod_{j=1}^k p_{m_j}$, $m \in \mathcal{C}^k$, is the stationary distribution of the embedded DTMC.

The proof is almost identical to that of [Theorem 2](#) and the same extensions hold for [Theorems 3](#) and [4](#) as well.

3.4 Variable Length Queue

We now consider a variable-length generalization. Unlike the model from [Section 2.2](#), the states can have different numbers of jobs present. The model is still *closed* in the following sense: after each completion, jobs arrive sequentially and are appended to the back of the queue until the state first hits a prescribed set $\mathcal{M}$ of valid states.

A state is a finite FCFS list $m = (m_1, \dots, m_{|m|})$ with entries in the class set $\mathcal{C}$ (finite or countable). Arrivals are i.i.d. with $\mathbb{P}(\text{arrival has class } c) = p_c$, where $p_c \geq 0$ and $\sum_{c \in \mathcal{C}} p_c = 1$. We fix a state space $\mathcal{M} \subseteq \mathcal{C}^* := \bigcup_{\ell \geq 0} \mathcal{C}^\ell$, whose elements may have unbounded length. We write $t \preceq m$ if t is a prefix of m , and define the prefix closure

$$\text{Pref}(\mathcal{M}) := \{t \in \mathcal{C}^* : \exists m \in \mathcal{M} \text{ such that } t \preceq m\}.$$

In state $x \in \mathcal{M}$, the next completion is chosen uniformly from positions $\{1, \dots, s(x)\}$, after which new jobs arrive i.i.d. according to $(p_c)_{c \in \mathcal{C}}$ until the state re-enters $\mathcal{M}$. We call this refilling.

We first impose assumptions ensuring that the refill step is well defined.

ASSUMPTION 3 (PREFIX-FREE $\mathcal{M}$). $\mathcal{M}$ is prefix-free: for any $m, m' \in \mathcal{M}$, we have $m \preceq m' \implies m = m'$.

ASSUMPTION 4. Let $A = (A_1, A_2, \dots)$ be i.i.d. with $\mathbb{P}(A_n = c) = p_c$. For each $t \in \text{Pref}(\mathcal{M})$, define the refill length as $\tau(t) := \inf\{n \geq 0 : (t, A_1, \dots, A_n) \in \mathcal{M}\}$. Assume that $\mathbb{P}(\tau(t) < \infty) = 1$ for all $t \in \text{Pref}(\mathcal{M})$.

ASSUMPTION 5 (PREFIX CLOSURE). For every $x = (x_1, \dots, x_{|x|}) \in \mathcal{M}$ and every $i \leq s(x)$, the list obtained by removing the i th job, $(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{|x|}) \in \text{Pref}(\mathcal{M})$.

DEFINITION 1. A variable-length queue is a queue satisfying [Assumption 3-5](#).

Two natural examples fit this variable-length framework. First, the color-based system described earlier, in which the server always serves the maximal front segment containing jobs of at most two different colors, naturally allows states of unbounded length. Second, the multiserver-job model with fractional server demands in $[0, 1]$ also fits this framework: the state may contain a variable number of jobs, and service is given to the maximal front segment whose total demand fits within capacity.

To guarantee a product form stationary distribution, we further impose a variable-length analogue of the fixed-length FIOI assumption ([Assumption 1](#)).

ASSUMPTION 6 (VARIABLE-LENGTH FIOI). For every $x \in \mathcal{M}$, if x' is obtained by permuting the first $s(x)$ entries of x , then $x' \in \mathcal{M}$ and $s(x') = s(x)$.

We consider the candidate product form $\pi(m) := \prod_{j=1}^{|m|} p_{m_j}$. A useful consequence of [Assumption 3](#) and [4](#) is a Kraft–McMillan type equality, which in particular implies that π is a valid probability distribution.

THEOREM 6. Under [Assumption 3](#) and [4](#), $\sum_{m \in \mathcal{M}} \pi(m) = 1$.

We now state the product form result for the variable-length queue.

THEOREM 7. In the variable-length queue, if [Assumption 6](#) holds, then $\pi(m) = \prod_{j=1}^{|m|} p_{m_j}$, $m \in \mathcal{M}$, is the stationary distribution of the embedded DTMC on $\mathcal{M}$.

4 Front Sum Order Independence

We now consider a saturated queue satisfying what we call the *Front Sum Order Independence* (FSOI) condition, motivated by the open-model literature of [\[4\]](#). For each position $i \in \{1, \dots, k\}$, let $\kappa_i : \mathcal{C}^i \rightarrow [0, \infty)$ denote the continuous-time completion rate of the job in position i , where κ_i depends only on the first i jobs in FCFS order. We set $\kappa_i(\cdot) = 0$ for all $i > s(m)$, so jobs outside service have zero completion rate.

For $j \in \{1, \dots, k\}$, write $K_j(u_1, \dots, u_j) := \sum_{i=1}^j \kappa_i(u_1, \dots, u_i)$ for the total completion rate of the first j positions, and in particular $K_k(m) = \sum_{i=1}^k \kappa_i(m_1, \dots, m_i)$. Assume $K_k(m) > 0$ for all m . The embedded DTMC then chooses the next completion position according to $\Delta_i(m) = \kappa_i(m_1, \dots, m_i) / K_k(m)$.

We further assume that the total completion rate of the front j positions is permutation invariant.

ASSUMPTION 7 (FSOI). For every $j \in \{1, \dots, k\}$, every $(u_1, \dots, u_j) \in \mathcal{C}^j$, and every permutation τ of $\{1, \dots, j\}$, we have $K_j(u_1, \dots, u_j) = K_j(u_{\tau(1)}, \dots, u_{\tau(j)})$. Equivalently, K_j depends only on the multiset of the first j classes.

Under FSOI, we obtain a different product form from the FIOI case.

THEOREM 8. Under [Assumption 7](#), the embedded DTMC has stationary distribution

$$\pi(m_1, \dots, m_k) = \frac{1}{G} \frac{\prod_{i=1}^k p_{m_i}}{\prod_{j=1}^{k-1} K_j(m_1, \dots, m_j)}, \quad m \in \mathcal{C}^k,$$

where G is the normalizing constant.

This is a DTMC result rather than a CTMC result. Accordingly, the K_k term does not appear in the denominator.

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